Course Description (IIIT-Delhi CSE course template)

Code	
Department	CSE
Name	Web Development for All
Credits	4
OfferedTo	UG
Description	This course provides a comprehensive grounding in full-stack web development principles, client-side dynamic interactivity, performance diagnostics, and secure backend service architecture. Students begin by constructing responsive multi-device web layouts using semantic HTML5 markup and CSS Flexbox and Grid models. They then progress to client-side scripting, utilizing asynchronous JavaScript, DOM manipulation, and Fetch APIs to build interactive web application features. The course further covers web performance profiling, where students diagnose network bottlenecks, rendering layout thrashing, and Core Web Vitals using browser developer tools. Finally, students evaluate RESTful backend service architectures with a focus on authentication patterns, OWASP security controls, and system scalability. Taught through 3 hours of lectures and 1 hour of tutorial per week over 13 weeks, the course enables students to build responsive front-end interfaces, integrate asynchronous web APIs, profile interface bottlenecks, and evaluate full-stack architectural trade-offs across usability, security, and scalability requirements.
Mandatory Prereg	None
Desired Prereq	None
Other Prereq	None
Antireq	None

Course Outcomes

CO #	Outcome	BL #
CO1	Write responsive web layouts using standard HTML5 markup and CSS styling rules for specified multi-device wireframes.	BL3
CO2	Implement dynamic client-side interactivity using asynchronous API requests and DOM manipulation for interactive web application features.	BL3
CO3	Diagnose performance bottlenecks and rendering issues in web interfaces with browser developer tools and performance metrics.	BL4
CO4	Evaluate RESTful backend service designs against security, scalability, and usability requirements for a full-stack web system.	BL5

2 of 4 COs target BL4–BL6 (template: at least half).

Approval Date	(new course: left empty)
Proposing Dept	CSE

Suggested Topics

List Of Topics	Relevant CO	Assignment/lab/tutorial	Average load (h/week)
Semantic HTML5 Markup and ARIA Accessibility Landmarks	CO1	Assign 1; Tutorial (week 1)	10
HTML5 Form Validation and Document Structure Rules	CO1	Assign 1; Tutorial (week 1)	10
CSS Flexbox Model and One-Dimensional Layout Alignment	CO1	Assign 1; Tutorial (week 2)	11
CSS Grid Systems and Two-Dimensional Container Layouts	CO1	Assign 1; Tutorial (week 2)	11
Responsive CSS Media Queries and Viewport Breakpoints	CO1	Assign 1; Tutorial (week 3)	12
DOM Manipulation Tree Operations and Event Listener Handling	CO2	Assign 1; Tutorial (week 4)	11

List Of Topics	Relevant CO	Assignment/lab/tutorial	Average load (h/week)
Asynchronous JavaScript, Promises, and Fetch API Integration	CO2	Assign 2; Tutorial (week 5)	12
Dynamic Client-Side Component Rendering	CO2	Assign 2; Tutorial (week 6)	12
Real-Time UI Feed Updates and Async Error Handling	CO2	Assign 2; Tutorial (week 6)	12
Browser Network Waterfall Profiling and Resource Timing Analysis	CO3	Tutorial (week 7)	10
JavaScript Execution Timelines and Main-Thread Layout Thrashing	CO3	Tutorial (week 8)	11
Core Web Vitals and Diagnostic Bottleneck Evaluation	CO3	Tutorial (week 9)	11
REST Architectural Constraints and Resource URI Conventions	CO4	Tutorial (week 10)	10
Web Application Authentication Strategies and OAuth/JWT Patterns	CO4	Report 1; Tutorial (week 11)	10
API Rate-Limiting Controls and OWASP Vulnerability Assessment	CO4	Report 1; Tutorial (week 11)	11
Backend Service Scalability and Load Balancing Architecture	CO4	Report 1; Tutorial (week 12)	11
Full-Stack Web Architecture Trade-off Evaluation	CO4	Report 1; Tutorial (week 12); Tutorial (week 13)	12

Average load = expected hours of weekly work outside lectures, including labs, tutorials, homework and self-study (template: 10–12 h for a 4-credit course). Mean over topics: 11.0 h.

Suggested Plan of Labs

Labs are not applicable to this course.

Suggested Assessment Plan

Type of Evaluation	% Contribution in Grade
Assignment (2)	25
Quiz (4, best 3) [proctored]	15
Mid-sem exam [proctored]	20
Report	10
End-sem exam [proctored]	30
Total	**100**

Proctored evaluation: 65% (template: at least 40%).

Suggested Resources

Name/Title	Type
Jon Duckett (2014). HTML & CSS. ISBN 9781118871645	Textbook
Marijn Haverbeke (2024). Eloquent Javascript. ISBN 9781718504110	Textbook
Ilya Grigorik (2014). High Performance Browser Networking. ISBN 9789351104711	Reference
Leonard Richardson, Michael Amundsen, Sam Ruby (2013). RESTful Web APIs: Services for a Changing World. ISBN 9781449358068	Reference
Dafydd Stuttard, Marcus Pinto (2011). The web application hacker's handbook. ISBN 9788126533404	Reference

CO mapping with the programmes offered by the CSE department

Cell = mapping strength (1 low, 2 medium, 3 high) with its explanation, as in the template's CO / CO-Explanation columns.

B.Tech. (common)	Program Outcome	CO1	CO2	CO3	CO4
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.				
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			**3** Diagnosing performance bottlenecks requires systematic analysis of network logs and browser execution metrics.	**3** Evaluating backend architectures involves analytical assessment of security and scalability constraints.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	**3** Writing responsive web layouts directly contributes to designing user interface components that satisfy multi-device functional specifications.	**3** Implementing dynamic client-side interactivity builds functional interactive application components.		
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.				
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.	**2** Utilizing modern standard HTML5 markup and CSS layout techniques requires applying current web development tools.	**2** Employs modern asynchronous fetch APIs and browser DOM manipulation mechanisms.	**1** Uses specialized browser developer tools and profiling instrumentation to diagnose rendering bottlenecks.	
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.				
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.				
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.				
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.				
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.				**2** Articulates architectural evaluations and design trade-offs within structured design documentation.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.				

B.Tech. (common)	Program Outcome	CO1	CO2	CO3	CO4
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.				

B.Tech.(CSE)	Program Outcome	CO1	CO2	CO3	CO4
PSO1	Understanding of theoretical foundations and limits of computing.				
PSO2	Understanding of computing at different levels of abstraction including circuits and computer architecture, operating systems, algorithms, and applications.				
PSO3	Ability to adapt established models, techniques, algorithms, data structures, etc. for efficiently solving new problems.		**2** Adapts established asynchronous programming patterns and event-handling techniques to implement dynamic client features.	**3** Analyzes bottleneck patterns using profiling data to identify interface rendering inefficiencies.	
PSO4	Ability to design, implement, and evaluate computer based system or application to meet the desired needs using modern tools and methodologies.	**3** Designing and implementing responsive frontend components fulfills application needs using contemporary web standards.	**3** Directly implements interactive client-side application modules communicating with backend endpoints.		**3** Evaluates full-stack architectural designs against specified non-functional requirements.
PSO5	Understanding and ability to use advanced techniques and tools in different areas of computing.	**2** Applies standard CSS layout engines and frontend markup techniques for multi-device web user interface engineering.		**1** Applies browser diagnostic toolchains and profiling techniques for web performance engineering.	

B.Tech.(CSAI)	Program Outcome	CO1	CO2	CO3	CO4
PSO1	Understanding of foundational topics in Computer Science, Artificial Intelligence, and Machine Learning.	**2** Requires basic foundational understanding of web technologies, markup structure, and document rendering principles.			
PSO2	Understanding of theoretical foundations and limits of Artificial Intelligence and Machine Learning.				
PSO3	Ability to design and implement algorithms and data structures for efficiently solving new problems.	**3** Translates wireframe layout specifications into structured DOM trees and responsive CSS rendering rules.	**3** Implements asynchronous data handling and DOM state update logic for interactive application workflows.		
PSO4	Ability to model and analyze a variety of problems using appropriate mathematical/computational and AI concepts.			**3** Analyzes quantitative network logs and rendering timeline data to pinpoint runtime bottlenecks.	**3** Analyzes web architecture documentation against systemic security and scalability trade-offs.
PSO5	Ability of apply and develop AI algorithms to transform large amount of data into intelligent decisions and/or behavior.		**3** Provides client-side interface integration mechanisms suitable for displaying data-driven application responses.		
PSO6	An understanding of the impact of AI based solutions in an economic, societal, and environment context.				

MTech(CSE)	Program Outcome	CO1	CO2	CO3	CO4
PO1	An ability to independently carry out research /investigation and development work to solve practical problems				
PO2	An ability to write and present a substantial technical report/document				**2** Produces written technical evaluations justifying architectural decisions against requirements.
PO3	Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program				
PSO1	An understanding of the theoretical foundations and the limits of computing.				
PSO2	An ability to adapt existing models, techniques, algorithms, data structures, etc. for efficiently solving problems.		**2** Adapts event-driven programming paradigms and asynchronous data fetch mechanisms to user interaction tasks.	**3** Applies performance diagnostic models to analyze empirical trace data and logs.	
PSO3	An ability to design, develop and evaluate new computer based systems for novel applications which meet the desired needs of industry and society.	**3** Developing responsive interfaces enables constructing compliant client-side presentation layers for modern applications.	**3** Develops interactive client-side execution layers integrated with RESTful application services.		**3** Critically evaluates server-side service designs to ensure robustness, security, and scalability.
PSO4	Understanding and ability to use advanced computing techniques and tools.	**2** Leverages HTML5 semantic elements and modern CSS layout mechanisms to implement multi-device web user interfaces.		**1** Utilizes advanced browser profiling and network diagnostic tools to evaluate interface performance.	
PSO5	An ability to undertake original research at the cutting edge of computer science & its related areas.				
PSO6	An ability to function effectively individually or as a part of a team to accomplish a state goal.				
PSO7	An understanding of professional and ethical responsibility.				
PSO8	An ability to communicate effectively with a wide range of audience.				
PSO9	An ability to learn independently and engage in lifelong learning.				
PSO10	An understanding of the impact of IT related solutions in an economic, societal, and environment context.				

Annex: week-by-week schedule

Week	Lecture topics	COs	Assessments
1	• Semantic HTML5 Markup and ARIA Accessibility Landmarks • HTML5 Form Validation and Document Structure Rules	CO1	
2	• CSS Flexbox Model and One-Dimensional Layout Alignment • CSS Grid Systems and Two-Dimensional Container Layouts	CO1	
3	• Responsive CSS Media Queries and Viewport Breakpoints • DOM Manipulation Tree Operations and Event Listener Handling	CO1, CO2	Quiz 1
4	• DOM Manipulation Tree Operations and Event Listener Handling • Asynchronous JavaScript, Promises, and Fetch API Integration	CO2	Assign 1 released
5	• Asynchronous JavaScript, Promises, and Fetch API Integration • Dynamic Client-Side Component Rendering	CO2	Assign 1 due Quiz 2

Week	Lecture topics	COs	Assessments
6	• Dynamic Client-Side Component Rendering • Real-Time UI Feed Updates and Async Error Handling • Browser Network Waterfall Profiling and Resource Timing Analysis	CO2, CO3	Assign 2 released
7	• Browser Network Waterfall Profiling and Resource Timing Analysis • JavaScript Execution Timelines and Main-Thread Layout Thrashing	CO3	Mid-sem exam
8	• JavaScript Execution Timelines and Main-Thread Layout Thrashing • Core Web Vitals and Diagnostic Bottleneck Evaluation	CO3	Assign 2 due
9	• Core Web Vitals and Diagnostic Bottleneck Evaluation • REST Architectural Constraints and Resource URI Conventions	CO3, CO4	Quiz 3
10	• REST Architectural Constraints and Resource URI Conventions • Web Application Authentication Strategies and OAuth/JWT Patterns • API Rate-Limiting Controls and OWASP Vulnerability Assessment	CO4	
11	• API Rate-Limiting Controls and OWASP Vulnerability Assessment • Backend Service Scalability and Load Balancing Architecture	CO4	Quiz 4
12	• Backend Service Scalability and Load Balancing Architecture • Full-Stack Web Architecture Trade-off Evaluation	CO4	Report 1 released
13	Project work and review		Report 1 due End-sem exam

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